To simplify Maths →

$$\boxed{\Sigma = \sigma^2 I}$$

when distribution is isotropic →

μ  Same variance in all direction

noisier image = something × noisy image + something else × noise

defined by us

q is chosen by us and its distribution is known (stochastic)

$$\boxed{q(x_t | x_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\, x_{t-1}, \beta_t I)}$$

Noiser image

Noise

Previous image

$$x_t = \sqrt{1-\beta_t}\, x_{t-1} + \sqrt{\beta_t}\, \epsilon \qquad \epsilon \simeq \mathcal{N}(0,1)$$

$$q(x_t / x_0) = \mathcal{N}(\sqrt{\bar{a}_t}\, x_0, (1-\bar{a}_t) I)$$

Clean image

$$\bar{a}_t = \prod_{i=1}^{t} a_i \quad \text{and} \quad a_t = 1-\beta_t$$

Learn Reverse process

] Learn to denoise

$\max_{\theta} \log p_{\theta}(x_0)$

training images

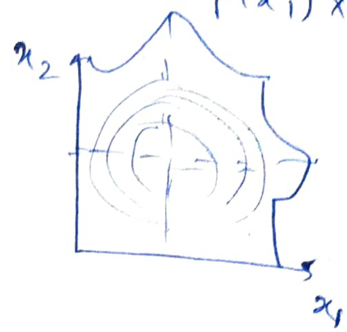
Probability given by the model of parameter θ

find θ

Joint PP

$P(x_1, x_2)$

$= p(x_1) \times p(x_2|x_1)$



Marginalization

$$p(x_1) = \underbrace{\int}_{\text{Sum across all } x} p(x_1, x_2)\, dx_2$$

$\log p_\theta(x_0) = \log \int P_\theta(x_0, x_{1:T})\, dx_{1:T}$

lower bound

ELBO (Evidence lower bound)

$$E_{x_0 \sim q(x_0)}\left[\log p_\theta(x_0)\right] \geq E_{x_{0:T} \sim q(x_{0:T})}\left[\log \frac{P_\theta(x_{0:T})}{q(x_{1:T}|x_0)}\right]$$

KL divergence

To check How far to PD is

$$KL(P\|Q) = \int \log\left(\frac{P(x)}{q(x)}\right) p(x)\, dx = E_{x\sim p}\left[\log\left(\frac{p(x)}{q(x)}\right)\right]$$

lower bound is tractable

$$P_\theta(x_{t-1}|x_t) = \mathcal{N}(\mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

Loss function $\Rightarrow$

$$\mathcal{L}_{DDPM} = E_{t, x_0, \epsilon}\left[\| \epsilon_\theta(\sqrt{\bar{a}_t}\, x_0 + \sqrt{1-\bar{a}_t}\,\epsilon, t) - \epsilon \|^2\right]$$

Because probability can be intractable we use log and then find gradient called as score matching

$$\boxed{S(x) = \nabla_x \log p(x)}$$

$$\nabla_x \log p(x) = \nabla_x \log f(x) - \cancel{\nabla_x \log Z}\ 0$$

$$\nabla \log(p) = \frac{\nabla p}{p}$$

when you follow the score (vector) you always go to region with highest density so we use langevin sampling

$$\boxed{x_t = x_{t-1} + \frac{a}{2} \nabla_x \log p_{data}(x_{t-1}) + \sqrt{a}\,\epsilon_t}$$

It return data from our probability distribution then.

Estimate score with $S_\theta(x)$

ISM ↓

$$\boxed{\mathcal{L}_{ISM} = E_x \left[\frac{1}{2} \| S_\theta(x) \|^2 + \nabla_x \cdot S_\theta(x) \right]}$$

SSM

$$\boxed{\mathcal{L}_{SSM} = E_{x_0, t, \epsilon, v} \left[2 v^T \nabla_x S_\theta(x) v + | v^T S_\theta(x) |^2 \right]}$$

But in Modern day we use different formula →

$$\boxed{S(x) = -\frac{(x - y)}{\sigma^2}}$$

$$\tilde{x} = x + \sigma\epsilon$$

so $q_\sigma(\tilde{x}|x) = \mathcal{N}(x, \sigma^2 I) \rightarrow \nabla_{\tilde{x}} \log q_\sigma(\tilde{x}|x) = -\frac{\tilde{x} - x}{\sigma^2}$

we then add noise in our data but maths is quite complicated so I am not doing this.

$$\boxed{q_\sigma(\tilde{x}) = \int \underbrace{q_\sigma(\tilde{x}|x)\, p_{data}(x)}_{q_\sigma(\tilde{x}, x)}\, dx}$$

when you do term interaction and expansion you get loss function.

$$\boxed{\mathcal{L}_{DSM} = E_{x;\tilde{x}} \left[\| S_\theta(\tilde{x}) - \underbrace{\nabla_{\tilde{x}} \log q_\sigma(\tilde{x}|x)}_{-\left(\frac{\tilde{x} - x}{\sigma^2}\right)} \|^2 \right]}$$

$\boxed{\sigma << 1}$ → cause poor distribution

multimodel-guided generation

forward

Difussion (DDPM)

$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon$

reverse

$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\, \epsilon_\theta(x,t)\right) + \sigma_t \epsilon$

$\epsilon_\theta(x,t)$ ← noise

Score match (SDE)

$dx = f(x,t)\,dt + g(t)\,dW$ ← noise

$dx = [f(x,t) - g(t)^2 s_\theta(x,t)]\,dt + g(t)\,d\bar{W}$

$s_\theta(x,t)$

Flow matching

$x_t = (1-t)x_0 + t x_1$

$x_{t+\Delta t} = x_t + u_\theta(x,t)\Delta t$

$u_\theta(x_t, t)$

1) Representation

Naive: Represent everything in pixel space

H, W → RGB $\boxed{n = H \times W \times 3}$ $x_{Im} \in \mathbb{R}^{H\times W\times 3}$

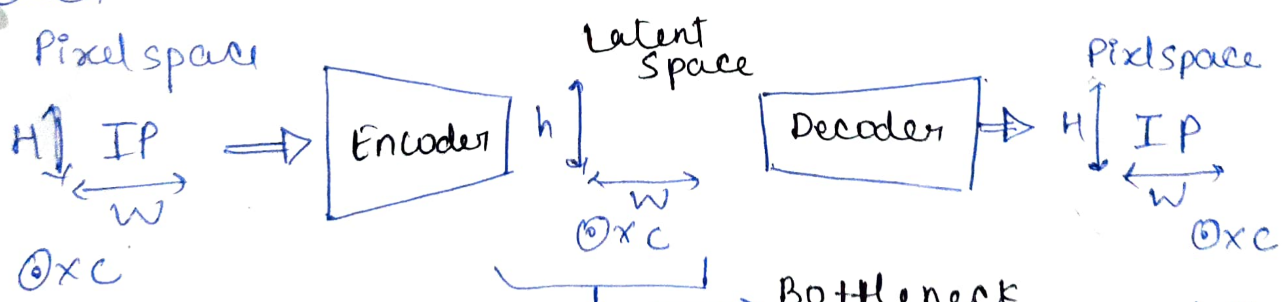
- High dimensionality issue.
- Redundant info
- Representation not meaningful.

what we want

- Tractable dimension
- compact representation
- Meaningful.

Attempt 1: via Autoencoders:

Arch



Job of AE is to find lower dimension representation (Latent) and then decode it

$$\boxed{\text{Spatial compression ratio} \quad f = \frac{H}{h} = \frac{W}{w}}$$

② Expand terms:

$$\boxed{ELBO = \underbrace{E_z[\log p_\theta(x/z)]}_{\text{Reconstructive term}} - \underbrace{KL(q_\varphi(z|x)\,||\,p(z))}_{\text{Regularize Latent space}}}$$

③ Loss function of VAE

$$\boxed{\underbrace{\mathcal{L}_{VAE} = -\mathbb{E}_z[\log p_\theta(x/z)]}_{\mathcal{L}_{rec}} + \underbrace{KL(q_\varphi(z|x)\,||\,p(z))}_{\mathcal{L}_{KL}}}$$

◦ Tractable • Compact ◦ meaningful • Truthful ×

In VAE There is issue of Bluriness.

Attempt 3: Adress limitation of VAE

$$\boxed{\mathcal{L}_{VAE} = \lambda_{rec}\mathcal{L}_{rec} + \lambda_{KL}\mathcal{L}_{KL}}$$

Refresher on reconstruction loss:

$$\mathcal{L}_{rec} = ||x - \hat{x}||^2$$

$\lambda_{rec} \simeq 1$ if high produce blurry loss.

on KL ⇃

$$\mathcal{L}_{KL} = KL(q_\varphi(z|x)\,||\,p(z))$$

$\boxed{\lambda_{KL} \simeq 10^{-6}}$ $\lambda_{KL}\uparrow$ lead to posterior loss.

Sol

Perceptual loss ⇃

If 2 images are slightly different not necessarily their feature map is different.

For that we use learned perceptual Patch similiarity captures Spatial difference

Representation of timestamp :→

Vector

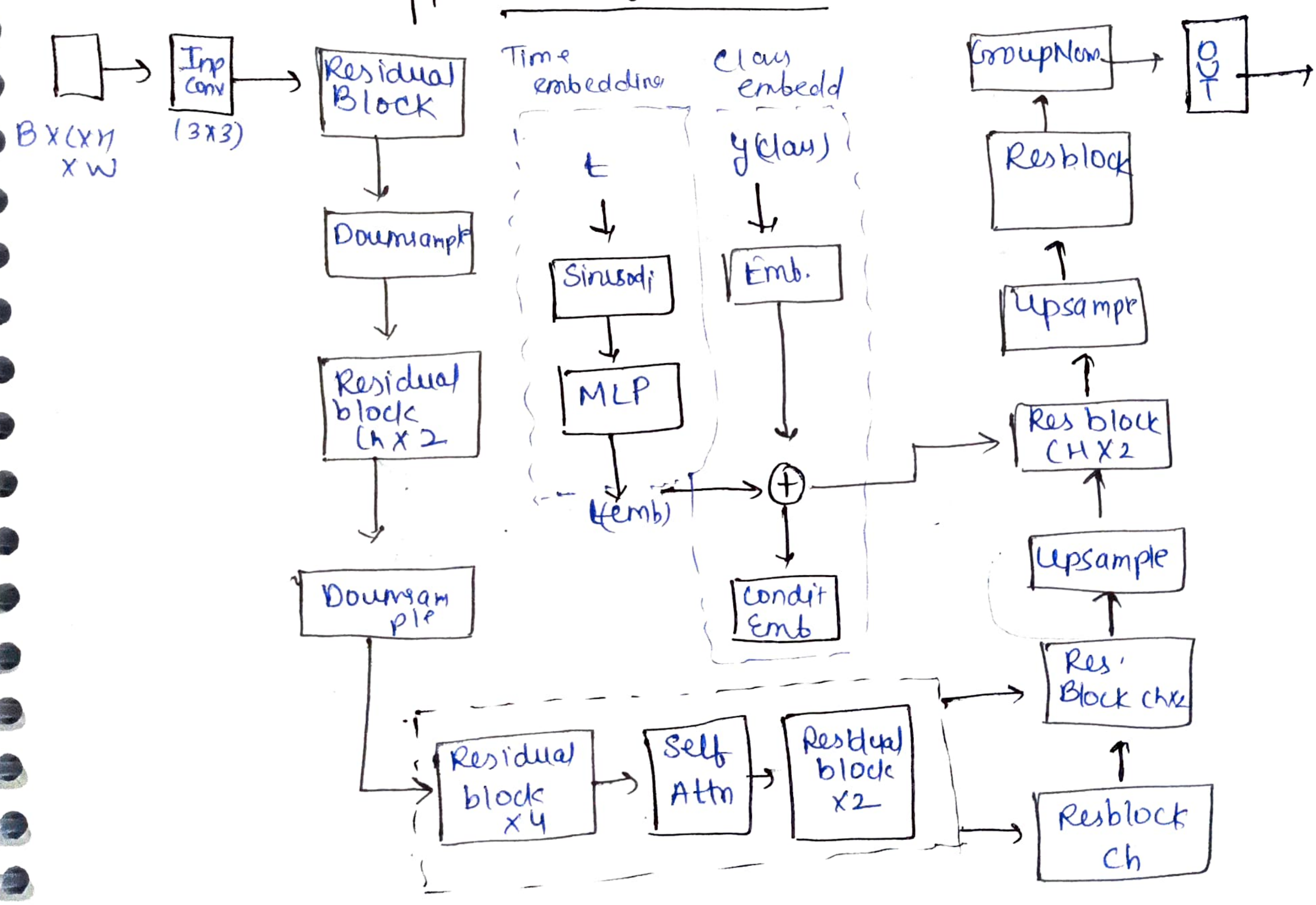
- Embedding corresponds to class lable
- Rich embeddings from predefined and LLM rich models

Adding condn info (t, c)

1: Add to featuremap
2: Modulate featuremap
3: Cross attention mechanism

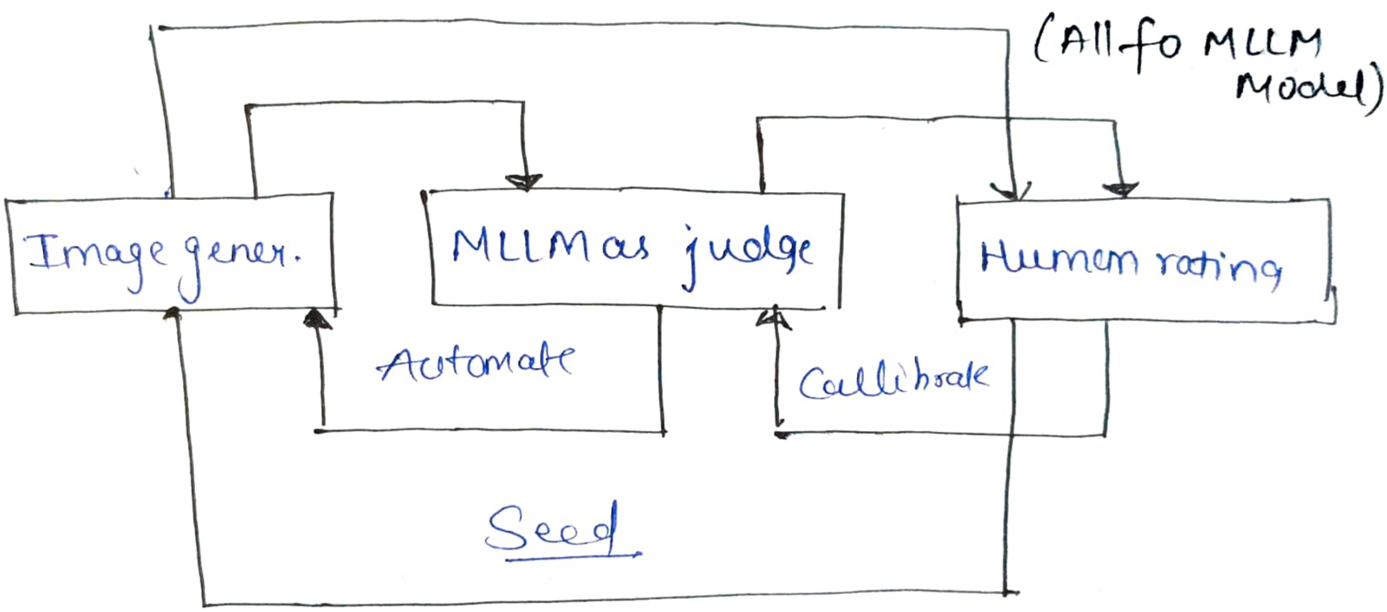
II)

Modified UNet



U-Net Architecture

- TIFA → (Text to image faithful Evaluation with QA)
  - ↳ Decompose faithfullness into Atomic Questions
- VQAScore = Visual Question Answering Score
  - ↳ [Image] Does this figure show [Prompt] ?
- VIEScore = Visual Instruction guided Explainable Score
  - ↳ MLLM as judge.



Workflow

Seed : Seek human expertise about task at it hand.
Callibrate : Allign MLLM-as-a judge with human expertise
Automate : Rely on MLLM-as-judge for general case